

useState

Exp 1

```
import { useState } from 'react';

export default function Exp1() {
  const [color, setColor] = useState("red");
  return (
    <>
      <h1>My favorite color is {color}!</h1>
      <div className="btns">
        <button type="button" onClick={() => setColor("blue")}>Blue</button>
        <button type="button" onClick={() => setColor("red")}>Red</button>
        <button type="button" onClick={() => setColor("pink")}>Pink</button>
        <button type="button" onClick={() => setColor("green")}>Green</button>
      </div>
    </>
  );
}
```

My favorite color is red!

Blue

Red

Pink

Green

Exp 2

```
import { useState } from 'react';

export default function exp2() {
  const [brand, setBrand] = useState("Ford");
  const [model, setModel] = useState("Mustang");
  const [year, setYear] = useState("1964");
  const [color, setColor] = useState("red");

  return (
    <>
      <h1>My {brand}</h1>
      <p>It is a {color} {model} from {year}.</p>
    </>
  )
}
```

My Ford

It is a red Mustang from 1964.

Exp 3

```
import { useState } from 'react';

export default function Exp3() {
  const [car, setCar] = useState({
    brand: "Ford",
    model: "Mustang",
    year: "1964",
    color: "red"
  });
}
```

```

    });

    return (
      <>
        <h1>My {car.brand}</h1>
        <p>It is a {car.color} {car.model} from {car.year}</p>
      </>
    )
  }
}

```

My Ford

It is a red Mustang from 1964.

Exp 4

```

import { useState } from 'react';

export default function Exp5() {
  const [car, setCar] = useState({
    brand: "Ford",
    model: "Mustang",
    year: "1964",
    color: "red"
  });

  const updateColor = () => {
    setCar(previousState => {
      return { ...previousState, color: "blue" }
    });
  }

  return (
    <>
      <h1>Exp 5</h1>
      <h1>My {car.brand}</h1>
      <p>It is a {car.color} {car.model} from {car.year}</p>
      <div className="btns">
        <button type="button" onClick={updateColor}>Blue</button>
      </div>
    </>
  )
}

```

My Ford

It is a red Mustang from 1964.

Blue

useEffect

Exp 5

```

import { useState, useEffect } from 'react';

export default function Exp4(){
  const [count, setCount] = useState(0);

  useEffect(() => {

```

```

        setTimeout(() => {
          setCount((count) => count + 1);
        }, 1000);
      });

      return (
        <>
          <h1>Exp 4</h1>
          <h1>I've rendered {count} times!</h1>
        </>
      );
    }
  }
}

```

I've rendered 155 times!

Exp6

```

import { useState, useEffect } from 'react';

export default function Esp6() {
  const [count, setCount] = useState(0);

  useEffect(() => {
    setTimeout(() => {
      setCount((count) => count + 1);
    }, 1000);
  }, []); // <- add empty brackets here

  return(
    <>
      <h1>I've rendered {count} times!</h1>
    </>
  );
}

```

I've rendered 2 times!

Exp 7

```

import { useState, useEffect } from 'react';

export default function Exp7() {
  const [count, setCount] = useState(0);
  const [calculation, setCalculation] = useState(0);

  useEffect(() => {
    setCalculation(() => count * 2);
  }, [count]); // <- add the count variable here

  return (
    <>
      <h1>Exp 7</h1>
      <p>Count: {count}</p>
      <div className="btns">
        <button onClick={() => setCount((c) => c + 1)}></button>
      </div>
      <p>Calculation: {calculation}</p>
    </>
  );
}

```

```
    );  
  }  
}
```

Count: 0



Calculation: 0

Exp 8

```
import { useState, useEffect } from 'react';  
  
export default function Exp8() {  
  const [count, setCount] = useState(0);  
  
  useEffect(() => {  
    let timer = setTimeout(() => {  
      setCount((count) => count + 1);  
    }, 1000);  
  
    return () => clearTimeout(timer)  
  }, []);  
  
  return(  
    <>  
      <h1>I've rendered {count} times!</h1>  
    </>  
  );  
}
```

I've rendered 1 times!

useContext

Exp9

```
import { useState, createContext, useContext } from 'react';  
  
const UserContext = createContext();  
  
export default function Exp9() {  
  const [user, setUser] = useState("Linus");  
  
  return (  
    <>  
      <UserContext.Provider value={user}>  
        <h1>{'Hello ${user}!'}</h1>  
        <Component2 />  
      </UserContext.Provider>  
    </>  
  );  
}  
  
function Component2() {  
  return (  
    <>  
      <h1>Component 2</h1>  
      <Component3 />  
    </>  
  );  
}
```

```

    );
  }

  function Component3() {
    const user = useContext(UserContext);

    return (
      <>
        <h1>Component 3</h1>
        <h2>`Hello ${user} again!`</h2>
      </>
    );
  }

```

Hello Linus!

Component 2

Component 3

Hello Linus again!

useRef

Exp10

```

import { useState, useRef, useEffect } from 'react';

export default function Exp10() {
  const [inputValue, setInputValue] = useState("");
  const count = useRef(0);

  useEffect(() => {
    count.current = count.current + 1;
  });

  return (
    <>
      <p>Type in the input field:</p>
      <input type="text" value={inputValue} onChange={(e) => setInputValue(e.target.value)} />
      <h1>Render Count: {count.current}</h1>
    </>
  );
}

```

Type in the input field:

Render Count: 0

Exp11

```

import { useRef, useState, useEffect } from 'react';

export default function Exp11() {
  const [inputValue, setInputValue] = useState("");
  const previousInputValue = useRef("");

  useEffect(() => {
    previousInputValue.current = inputValue;
  }, [inputValue]);
}

```

```

    return (
      <>
        <input type="text" value={inputValue} onChange={(e) => setInputValue(e.target.value)} placeholder='Type here text' />
        <h2>Current Value: {inputValue}</h2>
        <h2>Previous Value: {previousInputValue.current}</h2>
      </>
    );
  }
}

```

Type in the input field:

Current Value:

Previous Value:

useReducer

Exp 12

```

import { useReducer } from 'react';

const initialScore = [
  {
    id: 1,
    score: 0,
    name: "John",
  },
  {
    id: 2,
    score: 0,
    name: "Sally",
  },
];

const reducer = (state, action) => {
  switch (action.type) {
    case "INCREASE":
      return state.map((player) => {
        if (player.id === action.id) {
          return { ...player, score: player.score + 1 };
        } else {
          return player;
        }
      });
    default:
      return state;
  }
};

export default function Exp12() {
  const [score, dispatch] = useReducer(reducer, initialScore);

  const handleIncrease = (player) => {
    dispatch({ type: "INCREASE", id: player.id });
  };

  return (
    <>
      {score.map((player) => (
        <div key={player.id}>
          <label>

```

```
        <input type="button" onClick={() => handleIncrease(player)} value={player.name} />
        {player.score}
      </label>
    </div>
  )}
</>
);
}
```

John0

Sally0

useCallback

Without useCallback:

Count 1: 0

Count 2: 0

Button 1Button 2
